

AN ENGINEERING MODEL OF SPALLING FATIGUE FAILURE IN ROLLING CONTACT*

II. THE SURFACE MODEL

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(Received December 31, 1970)

SUMMARY

A model is advanced for explaining and predicting the incidence of rolling contact fatigue initiated at the contacting surfaces. The model includes fatigue initiating at (1) pre-existent surface discontinuities such as furrows and pits and (2) at the microgeometrical asperities on the surface.

The model accounts for the dependence of endurance life on the number and severity of surface defects, the film parameter (the ratio of the lubricant film thickness to the surface roughness), and the slope angle of the microgeometrical asperities. In addition, influence factors are provided to account for lubricant-metal surface interactions. None of these were explicitly covered in the existing theory of rolling bearing fatigue life prediction.

INTRODUCTION

Recent years have seen an increase in the frequency with which rolling bearings fail by a mode known as surface initiated spalling. The reason for this is that formerly the steel quality was so low that the likelihood of a bearing failing at a surface defect was negligible by comparison.

Present day improvements in steel making technology have made the frequency of incidence of surface and subsurface spalling comparable, with the consequence that surface initiated spalling has become much more common than in the Forties and earlier.

This explains why the Lundberg-Palmgren theory formulated in the Forties could successfully ignore surface spalling as a failure mode and why any modern theory for bearing fatigue life prediction must include it.

The present paper (the second of three papers published at this time) is an outgrowth of work done under the sponsorship of the Rome Air Development Center aimed at developing a mathematical model for bearing fatigue life prediction. It sets forth concepts arrived at in attempting to model surface initiated spalling. The first of

* Presented as an American Society of Lubrication Engineers paper at the ASME/ASLE Lubrication Conference held in Cincinnati, Ohio, October 13-15, 1970.

the three companion papers¹ describes the principles used in deriving this model and treats the subsurface case.

The *third* companion paper² interprets the basic life prediction equation in engineering terms and gives illustrative numerical examples of the influence of several physical variables on life.

Both the surface and subsurface models are postulated on the assumption that spalls are a consequence of the growth of cracks that emanate from localized defects. In the surface model these defects are divided into two classes:

Type B defects—indigenous defects such as furrows, digs and pits imparted to the surface in manufacture, or occurring early in bearing life from debris denting or microspalling at near-surface inclusions.

Type C defects—secondary micropits that form during running triggered by stress concentration at interacting surface asperities.

Subsurface defects are referred to as type A defects.

Figure 1 of ref. 1 is a logic diagram of the subsurface model. Figures 2 and 3 of that paper show the modifications of the general theory required for treatment of type B and type C defects respectively.

1. SURFACE FAILURE MODE A—FURROW ORIGINATED FAILURES

The steps in the development of a failure model for spalls originating at surface furrows or similar localized surface defects are quite similar to those used in the subsurface failure model¹.

As for subsurface defects it is assumed that the Phase I life N_1 of a rolling element that fails due to a spall that initiates at a furrow is given by

$$N_1 = C \left(\frac{\gamma_p}{D} \right)^{-\zeta} \quad (1)$$

where C , ζ are material constants and γ_p is the plastic strain at the furrow, D is the ductility at the furrow.

A difference between the surface and subsurface models arises in defining the micro stresses in the vicinity of the surface defect and another in determining the material fatigue strength constants since these should, in the case of surface originated failure, depend on the interaction between lubricant and rolling element material. While the question of microstress computation can be covered analytically as will be described presently, the question of how to describe lubricant interactions with the surface material is unresolved. It appears at this time that that will have to be approached experimentally. The model permits incorporation of experimental constants as they become available.

Computation of plastic strain at a furrow

A depression in the rolling surface of a rolling element can be an elongated one in the nature of a deep scratch, which is often parallel to the rolling direction and arises through certain manufacturing imperfections. (Such a scratch need not be deeper than 100×10^{-6} in. to be significant.) The depression can be a pit arising from exfoliation of a small area of surface material, or a dent created by plastic indentation of the surface, say, from a hard contaminant being rolled over. One can calculate the stress

raising effects of such imperfections from principles analogous to those used to calculate stress increases in the vicinity of subsurface inclusions. In ref. 3, the necessary elasticity calculations are described in detail. The geometry of the imperfection, considered a long and narrow "furrow" is illustrated in Fig. 1 which also shows that a pressure peak arises over the "shoulders" of the furrow as a result of the fact that no load is being carried over the furrow width.

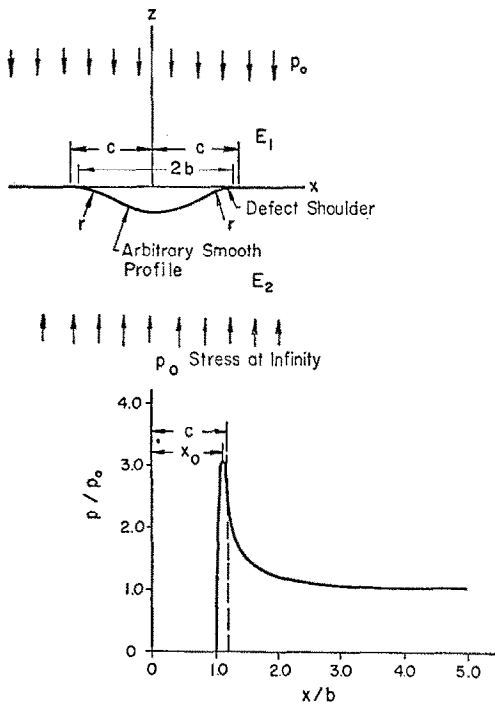


Fig. 1. Profile of surface furrow and contact pressure around a furrow.

It is found that two characteristics of the furrow define its severity as a stress raiser: the furrow width (or halfwidth) c and the furrow shoulder radius r (see Fig. 1). Given these parameters, the surface pressure p_0 that would exist at the furrow location in the absence of the furrow and the elastic moduli of the contacting surfaces, one can determine the magnitude of the pressure peak shown in Fig. 1 over the furrow shoulder. One can also analyze the subsurface shear stress arising in the material beneath the pressure peak, under the assumption that the material has infinite yield strength. Define a dimensionless furrow severity parameter⁴:

$$C_0 = \frac{\pi^2 p_0 r}{E' c} \quad (2)$$

Figure 2 taken from ref. 4 shows the relationship of the ratio between the maximum elastically calculated shear stress τ_c under the furrow shoulder and the undisturbed Hertzian pressure as a function of this parameter. By curve fitting to Fig. 2, eqn. (3) is obtained as a good approximation.

$$\frac{\tau_c}{p_0} = a_1 C_0^{a_2} \quad (3)$$

wherein $a_1 = 0.22$ and $a_2 = -0.24$.

One now proceeds to calculate the localized plastic strain as described in ref. 1 for subsurface defects. One significant difference in approach over that used for the subsurface model arises from the fact that the undisturbed alternating shear stress can be used therein to define stress and strain concentration factors, whereas there is no significant undisturbed alternating shear stress at the surface of a Hertzian contact.

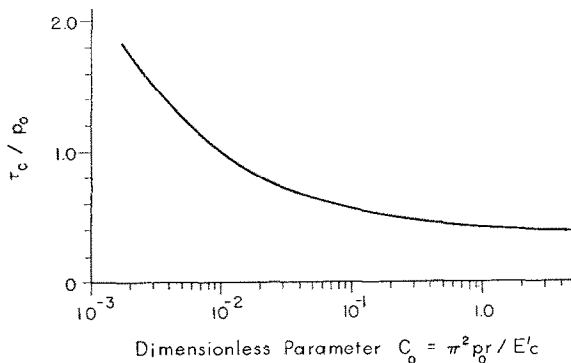


Fig. 2. Maximum shear stress vs. dimensionless parameter C_0 .

A slightly different treatment, based on the same principles, is therefore described in ref. 3 which utilizes the undisturbed Hertzian pressure p_0 to scale the stress concentration at the furrow. Let τ_1, γ_1 denote the actual shear stress and strain at the furrow. Using a stress-strain law proposed by Manson⁵ one has for any corresponding pair τ_1, γ_1 :

$$\gamma_1 = \frac{\tau_1}{2G} + \frac{3}{4}M \left(\frac{2\tau_1}{G_1} \right)^{z_1} \quad (4)$$

where M, z_1 and G_1 are material constants and G is the material's elastic shear modulus. In eqn. (4) the first right hand term is the elastic strain and the second term the plastic strain.

It is shown further in ref. 3 that to a good approximation one may write

$$\gamma_1 \tau_1 = \frac{\tau_c^2}{2G(1 + \sqrt{A/\rho})^2} \quad (5)$$

where A is the Neuber length (in.) and ρ is typical defect dimension analogous to notch radius in fatigue testing of notched specimens (in.)

For the purpose of calculating γ_p one must define the defect dimension ρ . The definition of ρ is that of a linear dimension of the order of magnitude of the stress concentration surrounding the defect. For a furrow, a logical dimension is the representative width of the pressure peak shown in Fig. 1. This distance ρ can be expressed in terms of the furrow geometry, and ref. 3 shows that the following approxi-

mate formula is satisfactory:

$$\rho \cong 14.4 c^{0.14} \left(\frac{p_0 r}{E'} \right)^{0.86} \quad (6)$$

It is noted that in this equation, the defect shoulder radius r has stronger influence on ρ than does the width of the furrow c .

For a given furrow geometry and pressure p_0 , one may calculate τ_c from eqns. (1) and (2) and thence γ_1 and τ_1 may be found by solving eqns. (3) and (4) simultaneously.

The plastic strain at the furrow γ_p is then calculated as the 2nd term on the right hand side of eqn. (4).

One proceeds by curve fitting to express γ_p as a function of the undisturbed Hertzian pressure p_0 in analogy to the treatment for subsurface inclusions. The expression thus found is:

$$\gamma_p = f'_2(\sigma_u) \cdot 10^{[7.1(\log(\tau^*/10.7))^{0.654} - 5.0]} \quad (6a)$$

where $f'_2(\sigma_u)$ = a function of hardness and

$$\tau^* = \frac{0.22(\pi^2 p_0 r / E' c)^{-0.24} p_0}{1 + \{(A/2c)(\pi^2 p_0 r / E' c)^{-0.86}\}^{\frac{1}{2}}} \quad (6b)$$

Ductility at a furrow

It is shown in ref. 3 that the relationship between ductility D and hydrostatic stress σ_h can be approximated by,

$$D = \frac{f_1(\sigma_u)}{(\sigma_h + 30,000)} \quad (7)$$

where $f_1(\sigma_u)$ denotes a hardness dependent function.

For a surface defect, σ_h is chosen as the nominal (undisturbed) hydrostatic stress at the defect location. σ_h is computed as the average of the normal stresses on three orthogonal surfaces i.e. $\sigma_h = \frac{1}{3}(\sigma_x + \sigma_y + \sigma_z)$. In an approximately two dimensional contact as in roller bearings and most ball bearings to a good approximation $\sigma_x = \sigma_z = p_0$ and $\sigma_y = \nu(\sigma_x + \sigma_z) = 2\nu p_0$ where ν is Poisson's ratio. Thus $\sigma_h = \frac{2}{3}(1 + \nu) \cdot p_0 \cong 0.9 p_0$ for steel.

Thus at a furrow one has approximately

$$D = \frac{f_1(\sigma_u)}{(0.9 p_0 + 30,000)} \quad (8)$$

Furrow life

Substituting from eqns. (6a) and (7) for γ_p and D into the Manson-Coffin life law of eqn. (1) gives a relatively complex equation:

$$N_I = C(\gamma_p/D)^{-\zeta} = C \left[\frac{f'_3(\sigma_u) \cdot 10^{[7.1(\log(\tau^*/10.7))^{0.654} - 5.0]}}{0.9 p_0 + 30,000} \right]^{-\zeta} \quad (9)$$

where $f'_3(\sigma_u) = f'_2(\sigma_u)/f_1(\sigma_u)$ is a hardness dependent function.

It has been clarified numerically that in eqn. (9), the ratio r/c is not a highly

influential variable. c , where it occurs in isolation, is, however, such an influential factor.

To proceed further it will be assumed that the ratio γ_p/D can be expressed as a power function in c and p_0 of the form

$$\frac{\gamma_p}{D} = f'_3(\sigma_u) c^{a_1} p_0^{b_1} \quad (10)$$

wherein a_1 and b_1 are constants assumed to be determined by curve fitting over the range of c and p_0 values of concern. If this is done the furrow life expression becomes

$$N_1 = C[f'_3(\sigma_u)]^{-\zeta} c^{-a_1\zeta} p_0^{-b_1\zeta} \quad (11)$$

The numerical work required to obtain this numerical approximation has not yet been accomplished, and in this respect, the furrow originated failure model is analytically incomplete.

Rolling body life

From eqn. (11), one proceeds through the same steps of probability analysis as discussed in ref. 1 for the subsurface failure mode, in order to obtain the failure life distribution of an entire rolling element. Of course, equi-hazard surfaces in the three dimensional subsurface volume are replaced by equi-hazard bands on the rolling contact surface, and the volume elements adjacent to an equi-hazard surface become strips of shallow depth extending along the rolling surface, down to a point where the shear stresses in the vicinity of furrows are no longer significant. Integration is then two-dimensional in determining the probability of rolling element survival.

In order to arrive at a final equation system one requires an assumption about the furrow width distribution $G(c)$. An assumption of the following form is shown in ref. 3 to be plausible for large c ,

$$m'[1 - G(c)] = \frac{\phi'_2}{c^{u'}} \quad (12)$$

wherein $G(c)$ is the probability of a furrow width $< c$, m' is the number of furrows per unit surface area ϕ'_2, u' are distribution parameters.

The literal meaning of the term ϕ'_2 is the expected number of defects per unit area exceeding the value $c=1.0$.

Generally, $G(c)$ will be obtained by curve fitting to furrow width data. It is necessary that $G(c)$ be determined by fitting to the high c portion of the distribution even if this means that there is substantial lack of fit in the low c range.

The reason for this practice is, of course, that the large c values are the ones which will result in failure so that the statistical description must be most valid for them.

If the fitted $G(c)$ does not agree with the data at small c values and if the units of c are chosen such that $c=1.0$ is a small value, ϕ'_2 will differ from the true expected number of defects per unit area greater than $c=1.0$.

Proceeding through a probabilistic reasoning such as described in ref. 1 with this assumed distribution form, the probability that the rolling element survives N_1 cycles without a furrow initiated failure is found to be expressible as,

$$-\log S(N_1) = \phi'_0 \phi'_2 \phi'_3 \phi'_4 N_1^{\mu'/a_1\zeta} \quad (13)$$

Wherein,

$$\phi'_0 = [Cf_3^{-1/5}(\sigma_w)]^{\mu'/a_1} \quad (14)$$

and

$$\phi'_3 \phi'_4 = \int_0^l \int_{-a}^a [p_0(x, y)]^{b_1 \mu'/a_1} dx dy \quad (15)$$

Wherein x, y are cross groove and circumferential coordinates, l is rolling body circumference, $p_0(x, y)$ is contact pressure p_0 as a function of x and y , $a = a(y)$ is semi-length of contact region.

For a Hertzian pressure distribution the integration with respect to x , may be carried out in dimensionless form. Let $p_{\max}(y)$ denote the maximum pressure at the cross section having circumferential coordinate y . Then one may write,

$$\phi'_3 \phi'_4 = \int_0^l \int_{-1}^1 [p_{\max}(y) \cdot \bar{p}]^{b_1 \mu'/a_1} d\bar{x} dy \quad (16)$$

Where $\bar{p} = p(x, y)/p_{\max}(y)$ and $\bar{x} = x/a$.

This reduces to

$$\phi'_3 = \int_{-1}^1 \bar{p}^{b_1 \mu'/a_1} d\bar{x} \quad (17)$$

$$\phi'_4 = \int_0^l a p_{\max}^{b_1 \mu'/a_1}(y) dy \quad (18)$$

where ϕ'_3 is independent of y . ϕ'_4 depends upon the circumferential rolling body load distribution. ϕ'_3 is a constant for Hertzian contacts irrespective of their size, but would assume other values in a non-Hertzian contact. (Therefore, in certain contact conditions ϕ'_3 may be load dependent and may vary around the rolling element circumference. In that case, only $[\phi'_3 \phi'_4]$ has meaning; the factors are not separable). Where separable, ϕ'_3 may be thought of as an axial stress distribution parameter. Notice that the analog to ϕ_1 in the surface model¹ does not appear in eqn. (13) since we have disregarded the equivalent of the defect severity factor K_t when neglecting the influence of r/c in approximating eqn. (9) by eqn. (10). Should it be desired to obtain greater sophistication, this influence can be reinstated and will result in the appearance of a ϕ'_1 factor.

The exponents a_1 and b_1 are unknown exponents resulting from the fact that eqn. (9) has not yet been fitted to a power law as required for eqn. (11).

It is mentioned that the effect of elastohydrodynamic films on the stress concentration in the vicinity of furrows has also been studied. The results are described in ref. 3, and it appears that as a first approximation, these effects are minor at least for film thickness values of the magnitude one can expect in rolling bearing operation. Stress peaks above the furrow shoulders form even in the presence of substantial elastohydrodynamic films, as a result of the fact that film pressure must drop to zero over the furrow. It is possible that more thorough analysis will later show that there is an elastohydrodynamic effect on the life of rolling elements failing at furrows, however, present theory does not suffice to predict it.

2. SURFACE FAILURE MODE B—SURFACE FATIGUE ORIGINATED SPALLING FAILURES

As described in ref. 6 a distinct mode of rolling contact spalling failure starts with the interaction of asperities through thin elastohydrodynamic films, resulting in the generation of a highly plastically worked ("glazed") surface layer, which through a fatigue mechanism, eventually forms a system of micro-cracks, resulting in a distribution of shallow ($< 100 \mu\text{in.}$) sharp-sided micropits over relatively large areas of the surface. Experience shows that a surface under rolling contact loading having achieved this micropitted condition, will then develop a spalling failure similar in appearance to that originating from pre-existing surface defects such as furrows, *i.e.* extending down to the region of high Hertz shear stresses. A reasonable assumption is that the micropits (or micro-crack systems) operate as secondary surface defects around which stress concentrations arise to generate the deep-growing cracks that form spalls. This set of phenomena is taken into account by computing the stress concentrations at asperity interactions and considering them the cause of the surface fatigue of which the micro-crack and micropit systems are the symptom. Recognizing that these microcracks and micropits do not lead to wholesale spalling of the entire surface, their formation is treated as a separate phase ("Phase 0") that ends with generalized micropitting on the surface. This micropitted surface is then treated as if it contained as many furrows, from which spalling fatigue failure arises according to the theory described in the previous section. Only sporadic facts are known about the precise time sequence of development of micropits, and about the variables influencing their generation. Therefore only the simplest model is justifiable. Such a simple model can be conceived using the hypothesis that the time required for the formation of the micropit system (Phase 0) is negligibly short by comparison to the life-time of the part to spalling failure. Accordingly, a life distribution for Phase 0 is not computed. Instead a set of probabilities will be given which describe the density and size distribution of micropits at the end of Phase 0, as a function of initial variables of the surface asperity distribution and of the lubrication condition.

In refs. 3 and 7, a simple elastic model is given for two straight-sided asperities with rounded tips coming into contact on the rolling surfaces. Figure 3 illustrates the model which is characterized by the rough surfaces being separated by an incompressible elastohydrodynamic lubricant film of mean thickness (between the center planes of the surfaces) h . In the places where the opposing asperities of heights H_1 , and H_2 and slope angles θ_1 , and θ_2 meet, the tips of the asperities are mutually being depressed to permit them to pass over each other. In this manner, a contact of width $2a$ is formed, and a total deformation equal to v , is suffered by the asperity tips. The lower half of Fig. 3 shows how one substitutes indentation of a rigid asperity into a smooth elastic surface for the situation just discussed. The substitution equations are:

$$\begin{aligned} H &= H_1 + H_2, \\ \theta &= \theta_1 + \theta_2, \\ E' &= \left[\frac{(1 - \nu_1^2)}{\pi E_1} + \frac{(1 - \nu_2^2)}{\pi E_2} \right]^{-1} \end{aligned} \quad (19)$$

where E_i , and ν_i denote the elastic modulus and Poisson's ratio of asperity i ($i = 1, 2$).

As described in ref. 3 one can calculate the maximum elastic shear stress τ_c , arising under the contact and the contact width a as a function of the ratio h/H of film thickness h and asperity height H . The parameters are: the asperity tip radius r , or the tip height δ_0 , and the asperity slope θ .

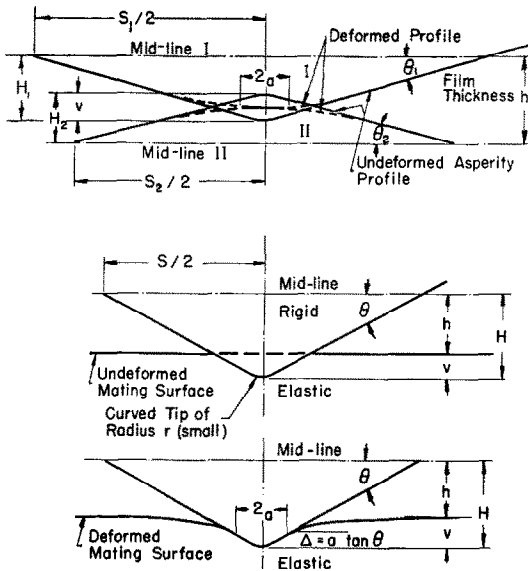


Fig. 3. Contact geometry of idealized asperities.

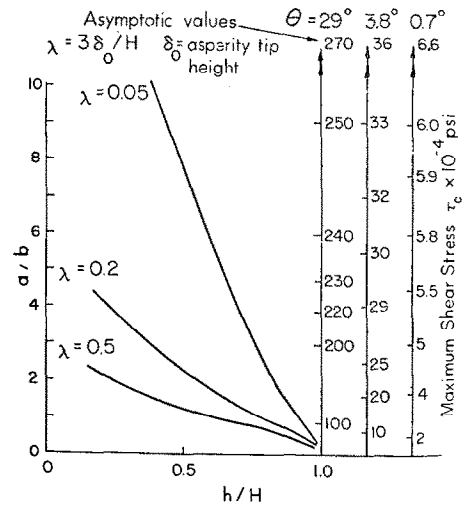


Fig. 4. Variation of maximum shear stress and a/b with h/H and λ for three types of surface finish (θ).

The calculation yields the set of approximate curves shown in Fig. 4 for various values of $\lambda = 3 \delta_0 / H$, with the contact width a scaled by the asperity tip width b (See ref. 7).

While the curves do not show it, due to the non-linear scale, τ_c has an asymptotic value for each θ , which has been annotated on each τ_c -scale. This asymptotic value is approached for *heavy* asperity interaction, i.e. low h/H or small tip radius compared to the width a and can be calculated as

$$\tau_{c,\infty} = \frac{E' \theta}{\pi^2} \quad (20)$$

The fact that the curves of Fig. 4 are approximately identical (with different ordinate scales for τ_c for all values of θ , indicates that θ is merely an ordinate scale factor. The factor is, from eqn. (20) the linear multiplier θ itself, i.e. for all values of h/H one has, approximately:

$$\frac{\tau_c(\theta, h/H)}{\tau_c(\theta_0, h/H)} = \frac{\theta}{\theta_0} \quad (21)$$

The asperity contact width a is, for small tip radii, independent of r , and can be obtained by solving from,

$$1 - \frac{h}{H} = \frac{2}{\pi} \left[\sin^{-1} (\Delta/H) + \frac{\Delta}{H} \log \left(\frac{2H}{\Delta} \right) \right]; \quad \Delta = a \tan \theta \quad (22)$$

A power function fit to this formula is

$$\frac{a \tan \theta}{H} \cong 0.6 \left(1 - \frac{h}{H} \right)^{0.8}, \quad h < H \quad (23)$$

Having obtained the asperity contact width a and stress τ_c , a hypothesis will now be constructed around these values to define the micropit distribution arising from them.

It is hypothesized that the pit generated by an asperity contact of width a is always of diameter $2c$ proportional to the contact width $2a$ (i.e. $2c = 2ka$ where the proportionality constant k depends upon the boundary lubrication condition at the asperity). The probability λ that such a pit forms is hypothesized to be dependent on the plastic microstrain severity parameter γ_p/D arising from the asperity contact. (In addition, an influence is exerted on λ , by the "boundary lubrication condition" at the asperity contact. This dependence will be made explicit below.) It is assumed that γ_p/D is obtainable from the elastic stress τ_c in a form similar to that used in eqn. (9) for the γ_p/D value for a furrow wherein, however, one uses for substitution into eqn. (9) values of τ_c from the curves in Fig. 4. It is recalled that the values of τ_c are dependent on tip radius and h/H and are of the form $\tau_c = (\theta/\theta_0)\tau_{c,0}$. For the limiting case of low h/H or small tip radius r , the value of $\tau_{c,\infty}$ is given by eqn. (20).

By analogy with eqn. (10) one can hope to fit a power function of the form

$$\gamma_p/D = f_3(\sigma_u)\theta^{a_1} [c(h/H)]^{a_1} \quad (24)$$

$c(h/H)$ signifies the functional relationships between c and h/H . Note that the pressure p_0 does not appear in this form, since the stresses are not dependent on p_0 , only on h/H .

The probability λ is

$$\lambda \left(\frac{\gamma_p}{D}, k' \right) = \lambda(\theta, c, \sigma_u, k') \quad (25)$$

wherein k' is another parameter that characterizes the boundary lubrication conditions and $c = c(h/H)$.

On each of the two rolling surfaces, there exists a distribution of asperity heights H (and tip radii r , to be considered a constant fraction of H) and slopes θ which are described statistically *e.g.* in refs. 8 and 9. The scale parameters of these distributions are the root mean square values of the asperity heights (the rms roughness σ) and the rms value of the slopes $\bar{\theta}$ respectively. The distribution of severity parameters γ_p/D will depend, in addition to $\bar{\theta}$ and σ on the mean elastohydrodynamic film thickness h . It is shown in ref. 7 that, for a given ratio h/σ the severity of the interactions depends only on slopes, *i.e.* $\bar{\theta}$. Since there are two rough surfaces in a rolling contact, asperity interactions arise from random encounters of opposing asperities and the parameters of both surface roughnesses enter the calculation. The simplest case, common in ball and some roller bearings, is that one of the surfaces is much smoother than the other, in which case only the asperities of the rougher surface are significant in determining interactions. Considering this case only one can deduce the distribution of asperity

contacts with a given value of contact half-width a and then, for given slope θ , the distribution of the severity parameters γ_p/D , from a knowledge of the asperity distribution on the rougher surface and of the elastohydrodynamic film thickness h .

One can compute the number of pits expected to arise over a unit area of rolled-over track by noting that the number of pits of diameter c equals the probability $\lambda(c)$ of a pit with diameter c , multiplied by the number of interactions of width $a = c/k$ per unit area.

The number of interactions having a width between a and $a + da$ per unit area is given by $vg(a)da$ where v is the total number of interactions of any size per unit area and $g(a)$ is the probability density function of the size distribution of interactions. If the unit area is large, v can be taken as a constant. The number per unit area of pits of size between $c = ka$ and $c + dc$ is

$$vg\left(\frac{c}{k}\right)\lambda(c)dc$$

The number of pits larger than some value $c_0 = ka_0$ is

$$v_p(c_0) = v \int_{c_0}^{\infty} g\left(\frac{c}{k}\right) \cdot \lambda(c) dc \quad (26)$$

Very little is known about $\lambda(c)$. It is plausible, however, to assume that

$$\lambda(c) = \lambda_1(k') \lambda_2(c)$$

where λ_2 does not depend on k' . This assumption implies that the effects of asperity slope and film thickness, combining to yield γ_p/D values, are independent of the boundary lubrication effects.

With this assumption, eqn. (26) becomes

$$v_p(c_0) = v \lambda_1(k') \int_{c_0}^{\infty} g\left(\frac{c}{k}\right) \lambda_2(c) dc \quad (27)$$

The distribution $g(a)$ can be approximated as a function of film thickness/roughness ratio h/σ if the statistics of surface asperities are known⁸.

From information given in ref. 9, a formula can be derived for $v(h/\sigma)$. It can be written

$$v = \frac{\pi}{2} N_0^2 \psi\left(\frac{h}{\sigma}\right) \quad (28)$$

where N_0 is the number of either-way crossings per unit length over the centerline of a profile tracing of the surface roughness and $\psi(h/\sigma)$ is a function of h/σ only, related to the error function, and is plotted in Fig. 5.

There is reason to believe that the integral in eqn. (26) is a function of h/σ of the type illustrated in Fig. 6, i.e. it goes to zero for high h/σ and is only weakly dependent on h/σ for lower values.

Since so little is known about $\lambda_1(k')$, one will assume that it is a constant $\lambda_1(k') = k' \leq 1$. If these assumptions hold, the function $v_p(h/\sigma)$ is, for any given c_0 and $h/\sigma < 3$, of the same form as $v(h/\sigma)$ i.e. it is illustrated in Fig. 5. For $h/\sigma \geq 3$, v_p goes to zero.

The distribution $v_p(c_0)$ comprises a description of the secondary surface defects (micropits) arising from asperity interactions, in the same terms as eqn. (12) is a des-

cription of the size distribution and density distribution of preexistent furrows. Life to spalling failure as determined by asperity interactions is then obtained from eqn. (13) assuming that the distribution can again be approximated by a form such as eqn. (12) as seems to be the case judging from experimental evidence of the Weibull distribution of surface fatigue initiated spalling failures.

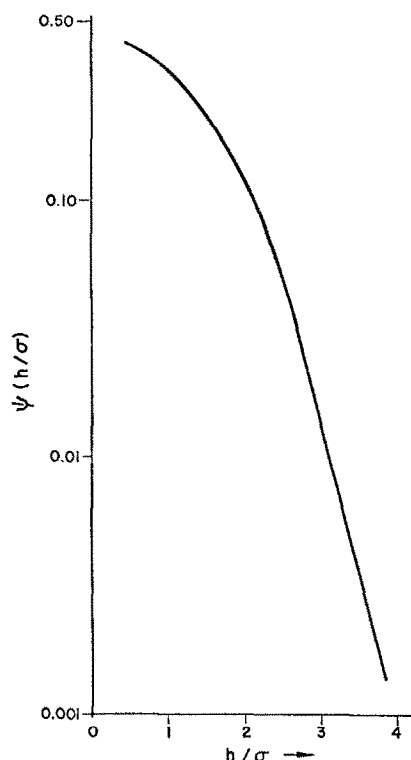


Fig. 5. The function $\psi(h/\sigma)$.

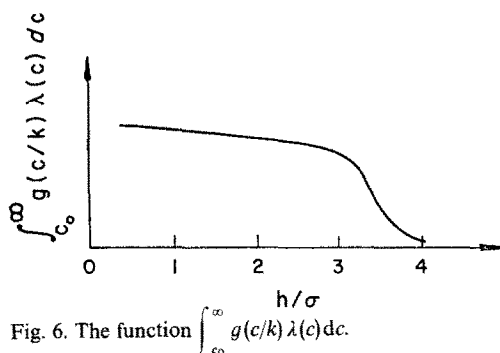


Fig. 6. The function $\int_{c_0}^{\infty} g(c/k) \lambda(c) dc$.

One can use Figs. 5 and 6 to construct an approximate relationship between ϕ'_2 of eqn. (12) and h/σ by noting in eqn. (12) that ϕ'_2 is the scale parameter of the distribution of defects. Since, from eqn. (27) $v_p(h/\sigma)$ is proportional to v , or from eqn. (28) to $\psi(h/\sigma)$, the scale parameter of the distribution $v_p(c)$ varies with h/σ as does $\psi(h/\sigma)$. These two statements, taken together, imply that

$$\phi'_2\left(\frac{h}{\sigma}\right) \sim \psi\left(\frac{h}{\sigma}\right), \quad \text{for } \frac{h}{\sigma} < 3. \quad (29)$$

For $h/\sigma > 3$, ϕ'_2 will drop below $\psi(h/\sigma)$ and with increasing h/σ become very small.

A curve of $\phi'_2(h/\sigma)$ has been sketched on these principles and is shown in Fig. 7.

The factor θ enters Fig. 6 by co-determining the details of the $h/\sigma < 3$ portion of the figure. No specifics of this effect are known but its direction is to increase the ordinate for higher θ .

For the prediction of Phase I life, in case more than one failure mode is operative, one must combine the failure probabilities considering all causes of failure. In the case of surface failure arising from furrows and from micropits formed by asperity interactions, this can be done by a suitable combination of the distribution of the two types of defects. The statistics of this combination are described in ref. 3. It is then, the combined distribution of surface defects of characteristic size c , that must be represented by eqn. (12) in order to obtain a life prediction from eqn. (13).

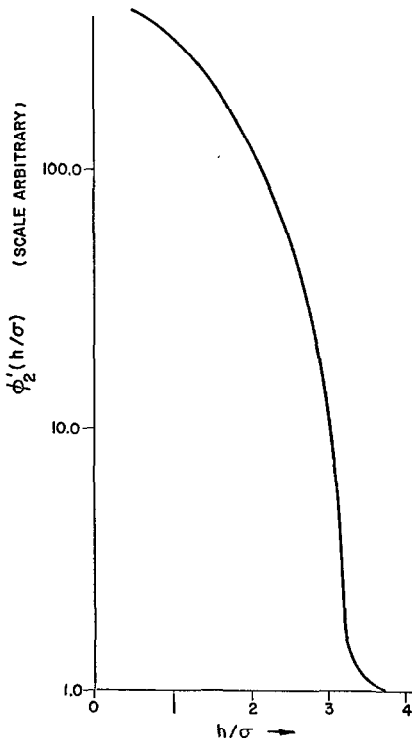


Fig. 7. The function $\phi'_2(h/\sigma)$.

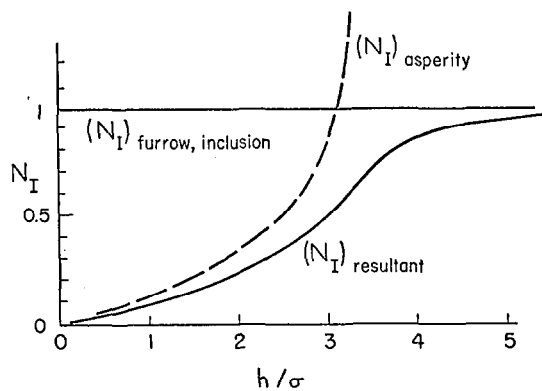


Fig. 8. Life vs. h/σ .

The dotted curve in Fig. 8 illustrates how the life to asperity originated spalling, *i.e.* any suitable percentile of the asperity life distribution, varies with the ratio (h/σ) of lubricant film thickness h to the composite roughness σ of the contacting surfaces.

The horizontal line shows the same percentile of the distribution of life due to a failure initiating at either a surface furrow or at a subsurface inclusion. Failure due to these modes is postulated to be independent of h/σ .

The resultant curve is the variation with h/σ of the same percentile of the distribution of life from any of the three failure modes.

This resultant curve is similar in general appearance to experimental observation.

ACKNOWLEDGEMENT

This paper is based on a study sponsored by Rome Air Development Center,

U. S. Air Force, under the project management of W. Bocchi. Support and permission to publish is gratefully acknowledged. The work was performed using the facilities of the Research Laboratory, SKF Industries, Inc., and numerous staff members have supplied valuable assistance. The authors are indebted for this collaboration.

NOMENCLATURE

A	= Neuber length,
a	= semi-major axis of the contact ellipse and semi-width of contact region of asperities,
$a_{1,2}$	= numerical constants,
C_0	= dimensionless furrow severity parameter,
c	= furrow width,
D	= ductility of matrix material,
E	= elastic modulus,
E'	= reduced Young's modulus,
$f_1, f_2, f_3(\sigma_u)$	= functions of hardness,
G	= material shear modulus,
$G(c)$	= probability of a furrow width c ,
G_1	= fatigue ductility constant,
$g(a)$	= probability density function of the size distribution of interactions,
H	= asperity height,
h	= elastohydrodynamic lubricant film thickness,
k	= proportionality constant,
k'	= a constant characterizing boundary lubrication conditions,
l	= rolling body circumference,
M	= fatigue ductility constant,
m'	= number of furrows per unit surface area,
N_1	= phase I defect life,
N_0	= number of either way crossings per unit length over the centerline of a profile tracing of the surface roughness,
p_0	= undisturbed Hertzian pressure,
r	= furrow shoulder radius,
τ	= shear stress,
τ_c	= maximum elastically calculated shear stress,
v	= deformation of Hertzian contact,
x, y	= cross groove and circumferential coordinates,
z_1	= fatigue ductility constant,
γ	= shear strain,
γ_p	= plastic shear strain range at defect,
δ_0	= asperity tip height,
ζ	= Manson-Coffin fatigue ductility exponent,
μ'	= defect dispersion parameter,
θ	= asperity angle,
$\bar{\theta}$	= rms asperity slope,
λ	= asperity parameter,
$\lambda(c)$	= probability of a pit with diameter c ,

- ν = total number of interactions of any size per unit area, Poisson's ratio,
 ρ = linear dimension of the order of magnitude of the stress concentration
 surrounding a defect,
 σ = rms roughness,
 σ_h = hydrostatic stress,
 σ_u = yield strength,
 $\phi_{1,\dots,4}$ = multiplicative modifying factors.

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